


```java
package com.cait.main;


import com.cait.student.*;
//import com.cait.student.Student;

public class StudentMain {

    public static void main(String[] args) {

        Student student1 = new Student();
        student1.setId(101);
        student1.setName("Abi");
        student1.setAge(22);
        student1.display();

        Student student2 = new Student();
        student2.setId(102);
        student2.setName("John");
        student2.setAge(23);
        student2.display();

        Student student3 = new Student();
        student3.setId(103);
        student3.setName("Priya");
        student3.setAge(21);
        student3.display();

        Student student4 = new Student();
        student4.setId(104);
        student4.setName("Rahul");
        student4.setAge(24);
        student4.display();

        Student student5 = new Student();
```

```
        student5.setId(105);
        student5.setName("Divya");
        student5.setAge(22);
        student5.display();

    }
}
```

```
package com.cait.student;

public class Student {

    private int id;
    private String name;
    private int age;

    // Getter and Setter for id
    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    // Getter and Setter for name
    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    // Getter and Setter for age
    public int getAge() {
        return age;
    }

    public void setAge(int age) {
```

```
        this.age = age;
    }

    public void display() {
        System.out.println("Id   : " + id);
        System.out.println("Name : " + name);
        System.out.println("Age  : " + age);
        System.out.println();
    }
}
```

```java
package com.stringmethods;

public class A_CharAt_Method {
    public static void main(String args[])
     {
          String name="Sachin";
          char ch1=name.charAt(2);//
          char ch2=name.charAt(11);
          System.out.println(ch1);
          System.out.println(ch2);
     }
}
```

```java
package com.stringmethods;

public class B_CompareTo_Method
{
    public static void main(String args[])
    {
     String s1 = "apple";
         String s2 = "apple";
         String s3 = "orange";
         String s4 = "apply";
         String s5 = "ant";

         //example 1
         System.out.println("Example 1: s1.compareTo(s2)");
         System.out.println("Output: " + s1.compareTo(s2));
// 0
         System.out.println("-----------------------------
----");

         //example 2
         System.out.println("Example 2: s1.compareTo(s3)");
         System.out.println("Output: " + s1.compareTo(s3));
// -14
         System.out.println("-----------------------------
----");
```

```
        //example 3
        System.out.println("Example 3: s1.compareTo(s4)");
        System.out.println("Output: " + s1.compareTo(s4));
// -20
        System.out.println("-----------------------------
----");

        //example 4
        System.out.println("Example 4: s1.compareTo(s5)");
        System.out.println("Output: " + s1.compareTo(s5));
// 2
        System.out.println("-----------------------------
----");
     }
}
```

```
package com.stringmethods;

public class C_Concat_Method
{
    public static void main(String args[])
    {
        String greet = "hello";
        greet.concat("world");
        System.out.println(greet);
        greet = greet.concat("world");
        System.out.println(greet);
    }
}
```

```
package com.stringmethods;

public class D_Equals_Method {

    public static void main(String args[]) {
        String s1 = "hello";
        String s2 = "hello";
        String s3 = "HELLO";
        String s4 = new String("hai");
```

```
        String s5 = new String("hai");


        System.out.println(s1.equals(s2));
        System.out.println(s1.equals(s3));
        System.out.println(s4.equals(s5));


        System.out.println(s4==s5); //false
        }
}
```

```
package com.stringmethods;
//CharSe
public class E_ContainsExample
{
    public static void main(String args[])
    {
        String name = "what do you know about me";
        System.out.println(name.contains("do you know"));
        System.out.println(name.contains("about"));
        System.out.println(name.contains("hello"));

        StringBuffer sb = new StringBuffer("do");

         System.out.println(name.contains(sb));


    }
}
```

```
package com.stringmethods;

public class F_EndsWith_Method {

     public static void main(String args[])
     {
          String s1="hello all";
```

```
            //StringBuffer sb = new StringBuffer("Hello");

            //System.out.println(s1.endsWith(sb));
            System.out.println(s1.endsWith("all"));
            System.out.println(s1.endsWith("l"));
            System.out.println(s1.endsWith("e"));


    }
}
```

```
package com.stringmethods;
//Diff SCP NSCP
public class G_Diff_SCP_NSCP {


    public static void main(String args[]) {
        String s1 = "hello"; //SCP
        String s2 = "hello"; //SCP


        String s3 = new String("hai");//NSCP
        String s4 = new String("hai"); //NSCP

        System.out.println(s1 == s2); //true
        System.out.println(s3 == s4);


    }
}

public class H_EqualsIgnoreCase {

    public static void main(String args[]) {
        String s1 = "hello";
        String s2 = "hello";
        String s3 = "HELLO";
        String s4 = "java";
        System.out.println(s1.equalsIgnoreCase(s2));
        System.out.println(s1.equalsIgnoreCase(s3));
        System.out.println(s1.equalsIgnoreCase(s4));
```

```
    }
}
```

```
package com.stringmethods;

public class I_IndexOf {

    public static void main(String args[]) {
        String s1 = "this is index of example";

        int index1 = s1.indexOf("is");
        System.out.println(index1);//2
        int index2 = s1.indexOf("index");
        System.out.println(index2);//8


        int index3 = s1.indexOf("is", 4);//5
        System.out.println(index3);



        //passing char value
        int index4 = s1.indexOf('s');//3
        System.out.println(index4);


        //passing char value
        int index5 = s1.indexOf('s',4);//6
        System.out.println(index5);
    }
}
```

```
package com.stringmethods;

public class J_IsEmpty_Method {

    public static void main(String args[]) {
        String s1 = "";
        String s2 = "hello";
```

```
        System.out.println(s1.isEmpty());
        System.out.println(s2.isEmpty());
    }
}
```

```
package com.stringmethods;

public class K_Length_Method {
    String test; // instance variable / non static
variable

    public static void main(String args[]) {
        String s1 = "java";
        String s2 = "python";

        String str = "Java8";
        String space = " ";

        System.out.println("string length is: " +
s1.length());
        System.out.println("string length is: " +
s2.length());

        if (str.length() > 0) {
            System.out.println("String is not empty and
length is: " + str.length());
        }
        str = "";
        if (str.length() == 0) {
            System.out.println("String is empty now: " +
str.length());
        }
        System.out.println("space length" +
space.length());

        K_Length_Method obj = new K_Length_Method();
        System.out.println("length of the test" +
obj.test.length());
    }
```

```
}
```

```
package com.stringmethods;

public class L_ReplaceMethod {

    public static void main(String args[]) {
        String s1 = "ha ha ha";
        String s2 = "My name is Khan. My name is Bob. My
name is Sonoo.";

        String replaceString = s1.replace('a',
'e');//replaces all occurrences of a char to e
        System.out.println(replaceString);
        String replaceString2 = s2.replace("My", "His");
        System.out.println(replaceString2);
    }
}
```

```
package com.stringmethods;

public class M_StringBufferUsage {

    public static void main(String args[]) {
       /*
        //Creating StringBuffer object
        StringBuffer test = new StringBuffer("good");
        System.out.println(test);
        */


        String str = "50000chars";
        StringBuffer stringBuffer = new StringBuffer(str);
        stringBuffer.append("dot");
        String newString = stringBuffer.toString();
        System.out.println("\n\nString is " + newString);

    }
}
```